Group Name **July 27, 2026**

Clarifier: Sharan
Manager: Zander
Reporter: Benjamin
Skeptic: Changhwan

Report: I am the reporter. I will remain nameless because of reasons. Someone in this group bought me boba which was really nice so I agreed to write the report. Today felt rather productive! We had pretty much all of our problem solving done by the time we had to go to the basement (sometimes called THE DUNGEON). So then we went and got boba. . . with permission of course. Anyways, that TA Rachel (Zander??), the really short TA is so cool. I hope to be as cool as them one day. Okay so that's all I got for now fam! Take it from someone who knows - it starts with the Boba - always starts with the Boba - next thing you know you are living in a basement in Califonia painting her house for no compensation and making eggs benedict from scratch for her family.

Rachel is Green. I am very flattered by your report!

SOLUTIONS & PROOFS:

Without Loss of Generality is abbreviated as WLoG within this document.

1. What is the fewest number of steps to go from 631452 to 123456?

   Solution. Our team has discovered two solutions of length 6. There are more methods available.

$$\begin{vmatrix}6\\3\\1\\4\\5\\2\end{vmatrix} \to \begin{vmatrix}2\\5\\4\\1\\3\\6\end{vmatrix} \to \begin{vmatrix}3\\1\\4\\5\\2\\6\end{vmatrix} \to \begin{vmatrix}5\\4\\1\\3\\2\\6\end{vmatrix} \to \begin{vmatrix}2\\3\\1\\4\\5\\6\end{vmatrix} \to \begin{vmatrix}3\\2\\1\\4\\5\\6\end{vmatrix} \to \begin{vmatrix}1\\2\\3\\4\\5\\6\end{vmatrix}$$

$$\begin{vmatrix}6\\3\\1\\4\\5\\2\end{vmatrix} \to \begin{vmatrix}1\\3\\6\\4\\5\\2\end{vmatrix} \to \begin{vmatrix}5\\4\\6\\3\\1\\2\end{vmatrix} \to \begin{vmatrix}4\\5\\6\\3\\1\\2\end{vmatrix} \to \begin{vmatrix}6\\5\\4\\3\\1\\2\end{vmatrix} \to \begin{vmatrix}2\\1\\3\\4\\5\\6\end{vmatrix} \to \begin{vmatrix}1\\2\\3\\4\\5\\6\end{vmatrix}$$

   We can refer to P_6 in order to discover the paths between the two states. This isn't optimal, but the optimal one is really hard to find! Good work.

2. Find an algorithm to take an arbitrary stack of pancakes and transform it into 12345 etc. The question is not asking for you to find the *fastest* way to do it, which is NP hard. I'm asking you to just find a method that always works. What is the most number of steps needed to order a stack with your algorithm?

   Solution. Let us explain by pseudocode. Let us us the basic repetitive functions along with FLIP(X) to flip right under X. Also let VAL(X) be the size of the pancake at the position X.

ALGORITHM SORT-PANCAKES
FOR ALL X IN ORDER FROM $(N \cdots 1)$:
$\rightarrow$ FLIP(N)
$\rightarrow$ FLIP(VAL(X)

Maybe a non-pseudocode explanation would actually help more here, but this algorithm works. How long does it take to complete?

3. Prove that the pancake graph for k pancakes is always regular. What is the degree of its vertices?

   Solution. We can know that the pancake graph is $k-1$ regular.

   Proof. For each permutation of pancakes, we have k naive flipping positions possible. (*SPATULA! YAY!*) However, since flipping at the topmost pancake changes nothing, we can ignore this move as invalid/unchanging. Therefore, each permutation of pancakes has $k-1$ valid moves, and each form an edge, therefore the graph is $k-1$-regular. Great! □

4. P_k is a path with k vertices. So P_1 is a vertex. P_2 is two vertices in a line, etc. For paths P_1 through P_9, classify them as N or P cutthroat positions.

   Solution. P, N, N, P, N, N, N, N, N Why? Why???? WHY???????????? (Sorry for the aggression. This is just the last homework I'm grading.) woooaahhhh, Zander crash out??!! The funny thing is, this is the first one I am grading so I haven't lost my marbles yet. I thought I found the marbles, but they were just those little balls for the boba tea

5. A "star" is the complete bipartite graph $K_{1,n}$. It looks like a $\star$. The n is the number of petals, and the 1 is the number one. A star is *odd* if it has an odd number of petals. I probably think you know what even means.

   (a) Classify the graph $K_{1,10} + K_{1,8}$ as N or P. Go ahead and prove your answer.

   Proof. If N were to remove a 'nova' from a graph, P would then remove a nova from another graph; therefore, N must remove a 'petal'. By the same logic, P must also remove a petal. If only a single petal remains in a graph, neither N nor P can remove it (as doing so would allow the opponent to remove a nova from another graph and win). Consequently, both players will continue removing petals from the graphs until only one petal remains in each; in this scenario, P wins. It can be applied all of $K_{1,2k}, K_{1,2l}$ graphs. Great! □

   (b) Classify the graph $K_{1,5} + K_{1,6}$ as N or P. Go ahead and prove your answer.

   Proof. If N removes one 'petal' from $K_{1,5}$, the situation becomes identical to that of $K_{1,4}$ or $K_{1,6}$, and the players' roles are swapped. In this scenario with swapped roles, the player who has taken on the new role wins; consequently, N emerges victorious. Great! □

   (c) Classify the graph $K_{1,4} + K_{1,5} + K_{1,6}$ as N or P. You don't have to prove your answer.

   Proof. If N removes the $K_{1,5}$ nova, the situation becomes identical to the case we just examined. Therefore, N wins in this instance as well. Great! □

(d) Prove that a game of cutthroat on a collection of stars is a P game IFF there are an even number of odd stars and an even number of even stars.

Proof. ($\Leftarrow$) We will prove this by using induction. base step: Nothing exist. Nothing is even. I concur.Obviously P game. induction step 1: If G are P game, then $G + K_{1,2k+1} + K_{1,2l+1}$ also P game because if N plays in G, P can plays in G by its own strategy. Or if N plays in $K_{1,2k+1}$ or $K_{1,2l+1}$ then P plays proper way to beat N. uh what do you mean by "the proper way to beat N"induction step 2: If G are P game, then $G + K_{1,2k} + K_{1,2l}$ also P game because by the same reason right above. What about the other direction?Yeah! What are you HIDING? □

6. Let G be a graph. Prove $G + G$ is a P game. *Alternative : Prove that Petersen + Petersen is a P game.*

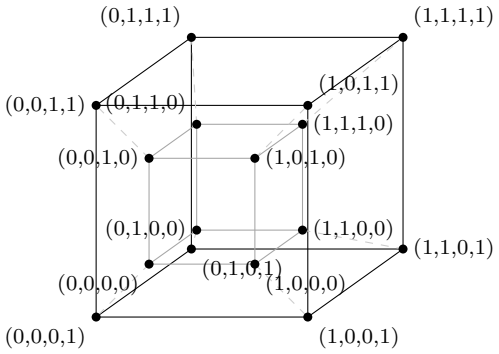
Proof. We will prove that the player P will **ALWAYS** be able to mimic the player N's actions in other graph. Then, the two graph G_1, G_2 will always maintain the same shape after each player P's actions. If one of graph G_1 or G_2 doesn't have any vertex to eliminate after P's action, then other one also doesn't have that vertex too. So I think you've hinted at the correct argument here (mimic N's move) but the way you've stated it, it sounds like you haven't actually done the argument. □

7. Classify paths beyond P_9 as N or P. Do you have any conjectures?

Conjecture. All of $P_k, (k > 9)$ are N because $P_5 \cdots P_9$ are N graphs. P_{10} has been found to be N as well.You should probably compute a few more examples to be sure; checking only up to 9 isn't even enough to realize that numbers can have more than one digit!

8. Is the Hypercube Graph(Q_4) N or P? Make sure to prove your answer.

Solution. P. Player P can independently imitate player N's actions via reflecting the player N's movements on the centroid of the hypercube. Mathematically, if we define X Y Z on the space and O as the layer(inner/outer) from the (X,Y,Z,O) coordinate tuple, we can flip the X Y Z bits in order to find our target move for P.



I genuinely wanna know how you have the patience to code this . . . cool stuff!

Proof. Assume not. Then, there is a move for N that P cannot imitate by flipping the X Y Z bits. Let the vertex removed by N be A and its reflection be B. This means that there is at least one neighbor of A that has not been removed till now, and all the neighbors of B have been removed.

WLoG take $A = (0,0,0,1)$, $B = (1,1,1,1)$. The neighbors of A are $(0,0,1,1)$, $(1,0,0,1)$, and $(0,1,0,1)$, while the neighbors B are $(0,1,1,1)$, $(1,0,1,1)$, and $(1,1,0,1)$. We can notice that there is a one to one function(a reflection relation) between the three neighbors of A and B. Thus, since all of B's neighbors were removed, all of A's neighbors should also have been removed.

Contradiction. Therefore P can always imitate N's movements and therefore can always win. Great! This works.Very nice proof □

9. Prove by induction that every connected graph of order at least two has at least two vertices, a and b, such that $G - a$ is connected, and $G - b$ is connected.

   Proof. Let us use proof by induction.

   (a) Base Case $n = 2$.
   Both vertices act as a and b, and removing either one leaves a singular and connected vertex.

   (b) Inductive Step.
   Assume we have a graph G_k of order $k \geq 2$ that is connected and containing vertices a_k and b_k such that $G - a_k$ and $G - b_k$ are connected.
   We add a new vertex v_{k+1} to produce G_{k+1}. There's a subtle error here; you're effectively assuming that your order $k+1$ graph G_{k+1} contains a vertex whose removal leaves the graph connected. This already assumes half of what you're trying to prove!!We simply assign v_{k+1} as a_{k+1} (WLoG), and one of a_k or b_k as b_{k+1}. This works because if v_{k+1} is only connected to either a_k or b_k, we choose the other as b_{k+1}.

   □

10. *CHALLENGE.* Assume the result of problem 5d is proven. Prove or disprove : Let s be the number of stars in a game of cutthroat on a collection of stars. Let the number of petals on star $j = p_j$. Let $M = s + \sum_{j=1}^{s}(p_j - 1)$. If M is even, the collection is a P game. If M is odd, the collection is an N game.

11. *CHALLENGE.* Is the petersen Graph N or P? Make sure to prove your answer.

    We guess N. That's rough, buddy.

12. *UNSOLVED.* Tree Packing Conjecture made by Gyárfás: Any set of trees with $2, 3, \ldots, k$ vertices has an (edge=disjoint) packing into the complete graph on k vertices.

13. *SOLVED?* How can one class be so awesome. **Answer.** Two conditions: have awesome people (✓) and have an awesome subject (✓). Very nice! Thank you for responding :)

14. *UNSOLVED.* As $n \to \infty$, does the pancake number P_n approach $\frac{3}{2}n$, $\frac{1+\sqrt{5}}{2}n$, or something else?

    Hehehehehe, I am chuckling

Nicely done overall!



Figure 1: ***Pancakes***